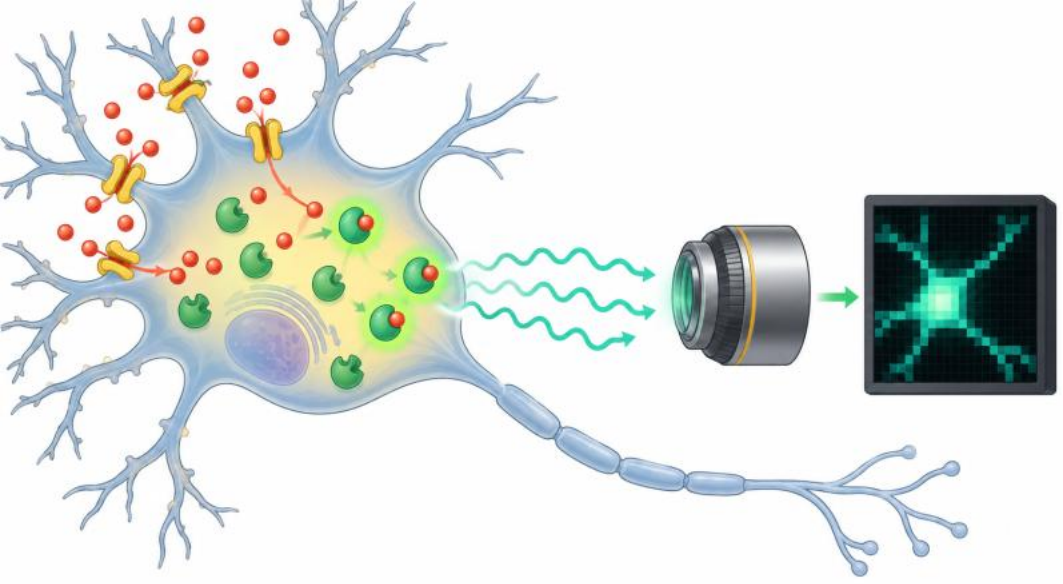
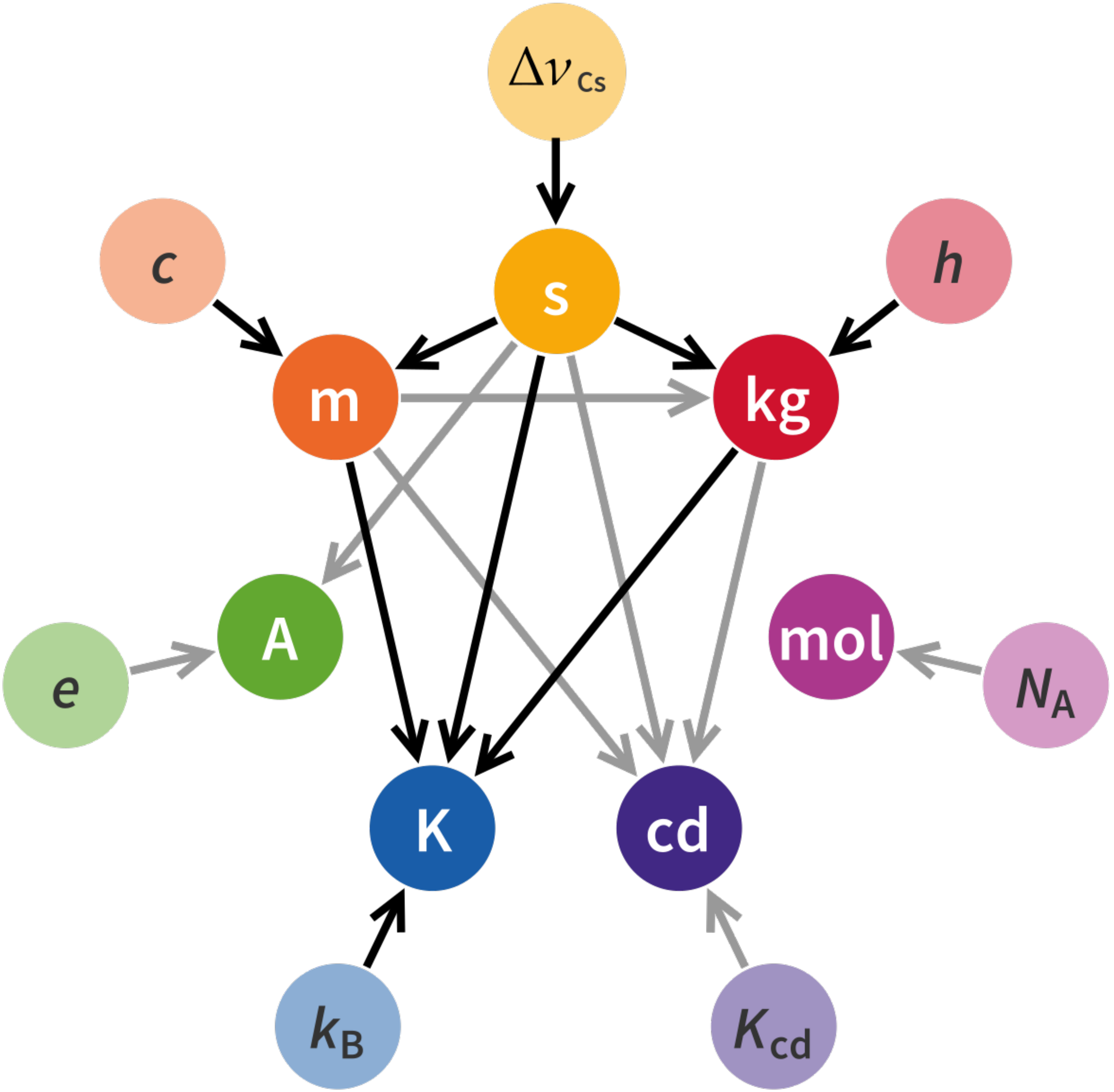
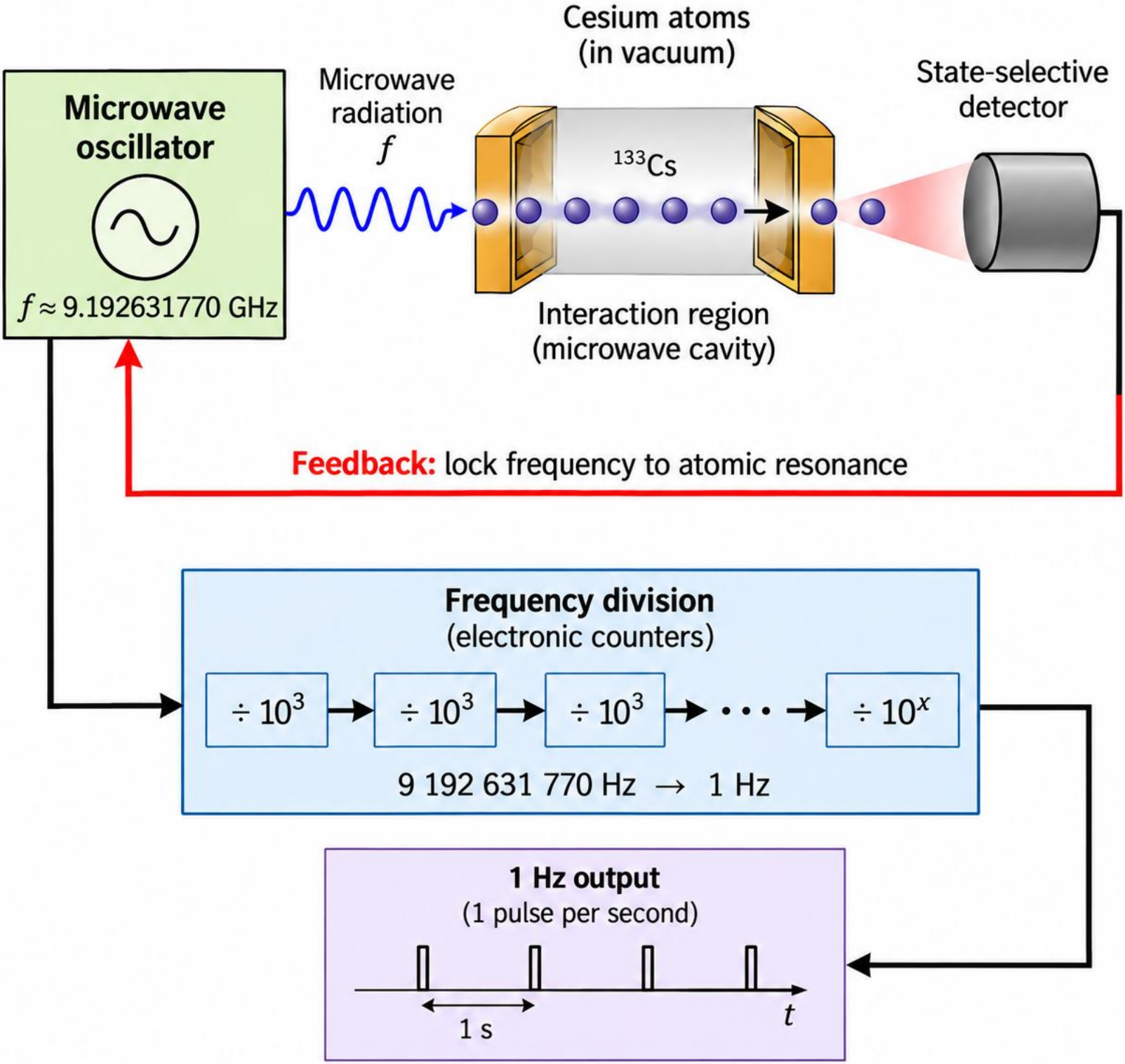
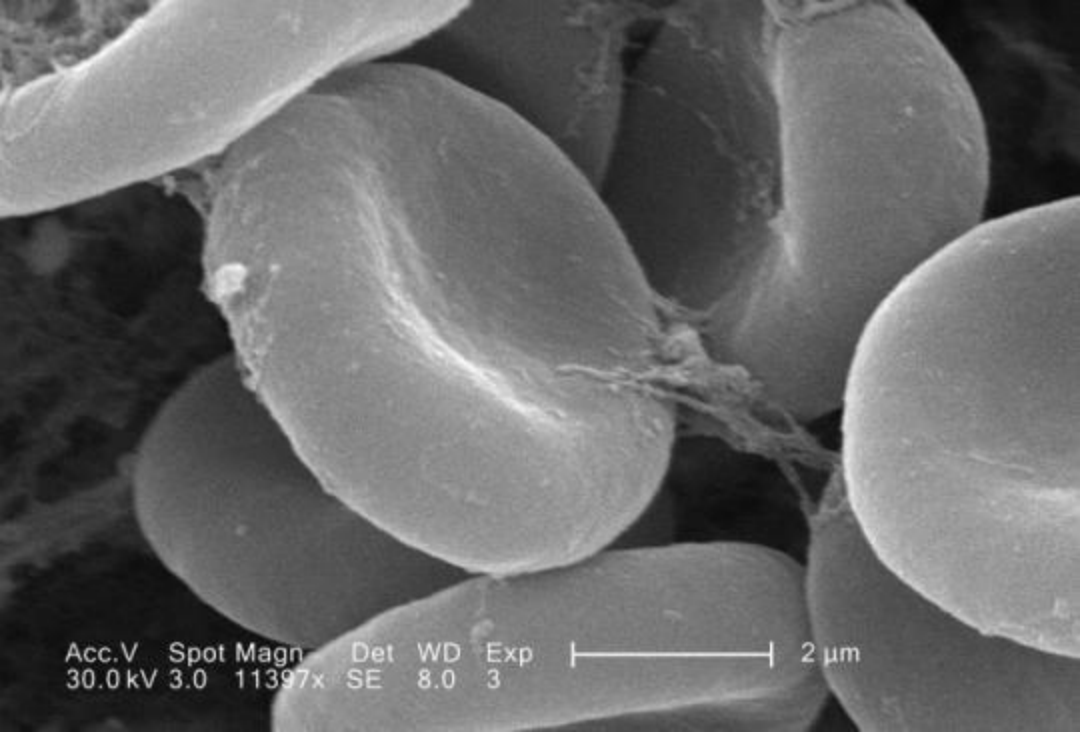
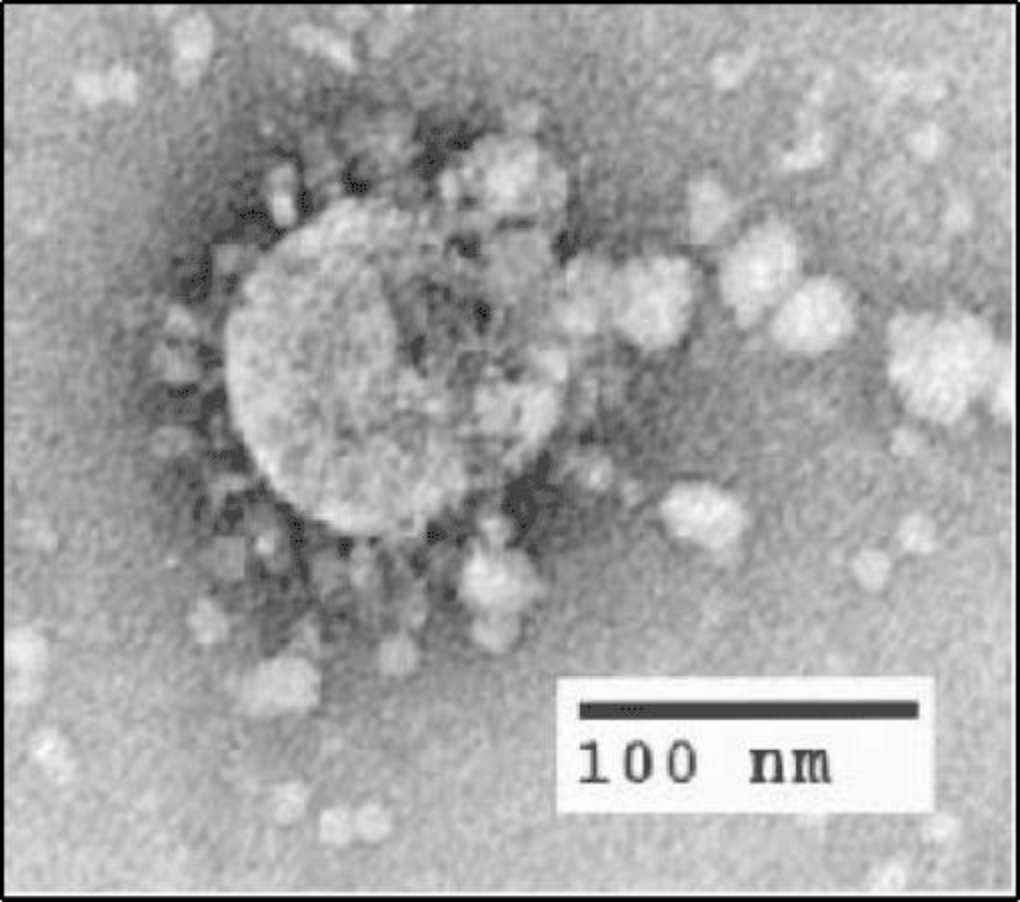

TARGET QUANTITY
Quantity intended to be measured
- Question: “What do I want to know?”
MEASUREMENT
Experimental process that produces information about the measurand
- Question: “How do I determine it?”
RESULT
Value attributed to the measurand
Reported with unit and uncertainty

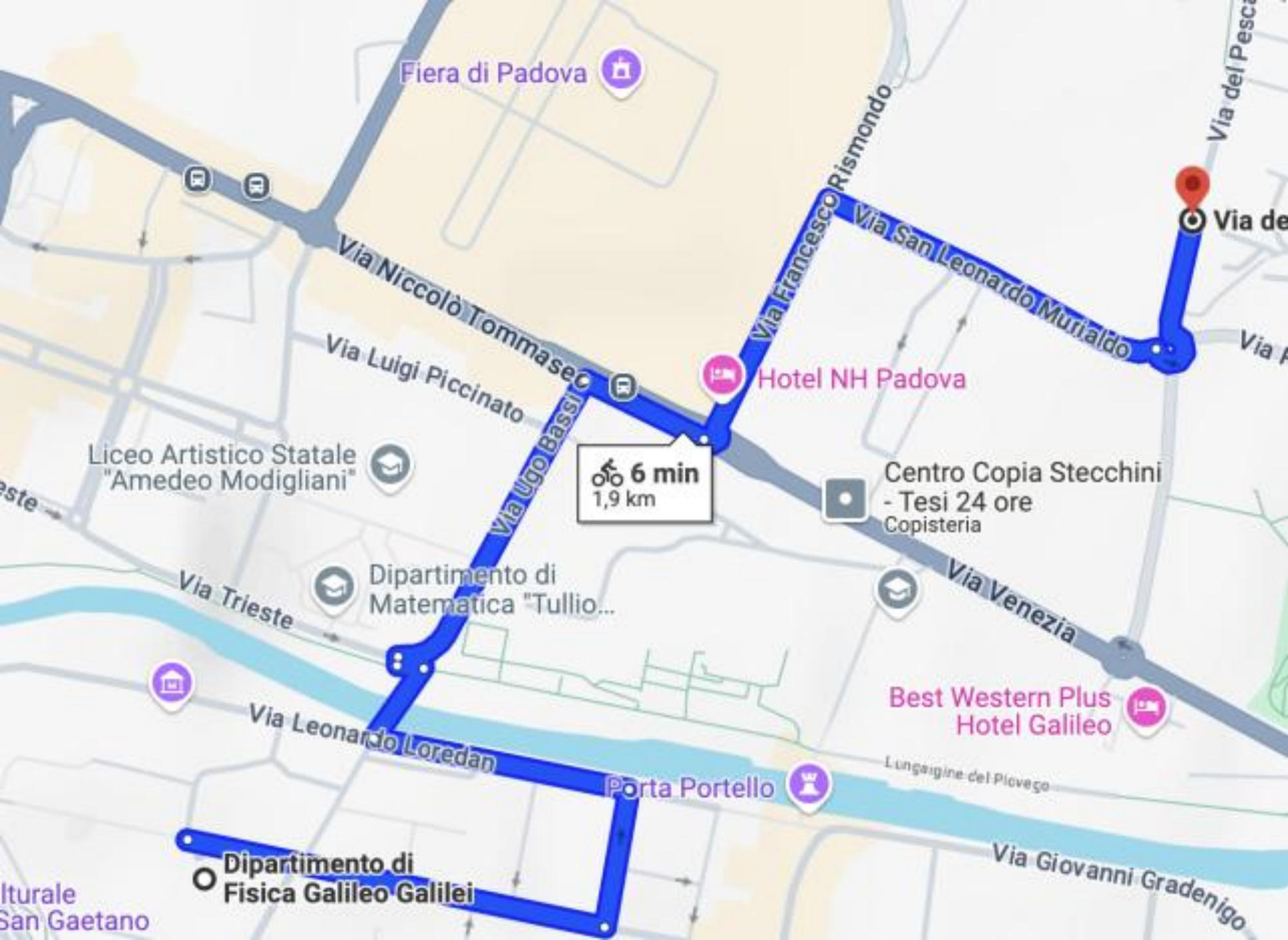



$\Delta\nu_{Cs}$
c
s
h
m
kg
e
A
mol
N_A
K
cd
k_B
K_{cd}


Cesium atoms
(in vacuum)
Microwave oscillator
Microwave radiation
f
State-selective detector
^{133}Cs
$f \approx 9.192631770$ GHz
Interaction region
(microwave cavity)
Feedback: lock frequency to atomic resonance
Frequency division
(electronic counters)
$\div 10^3$
$\div 10^3$
$\div 10^3$
$\div 10^x$
9 192 631 770 Hz → 1 Hz
1 Hz output
(1 pulse per second)
1 s
t

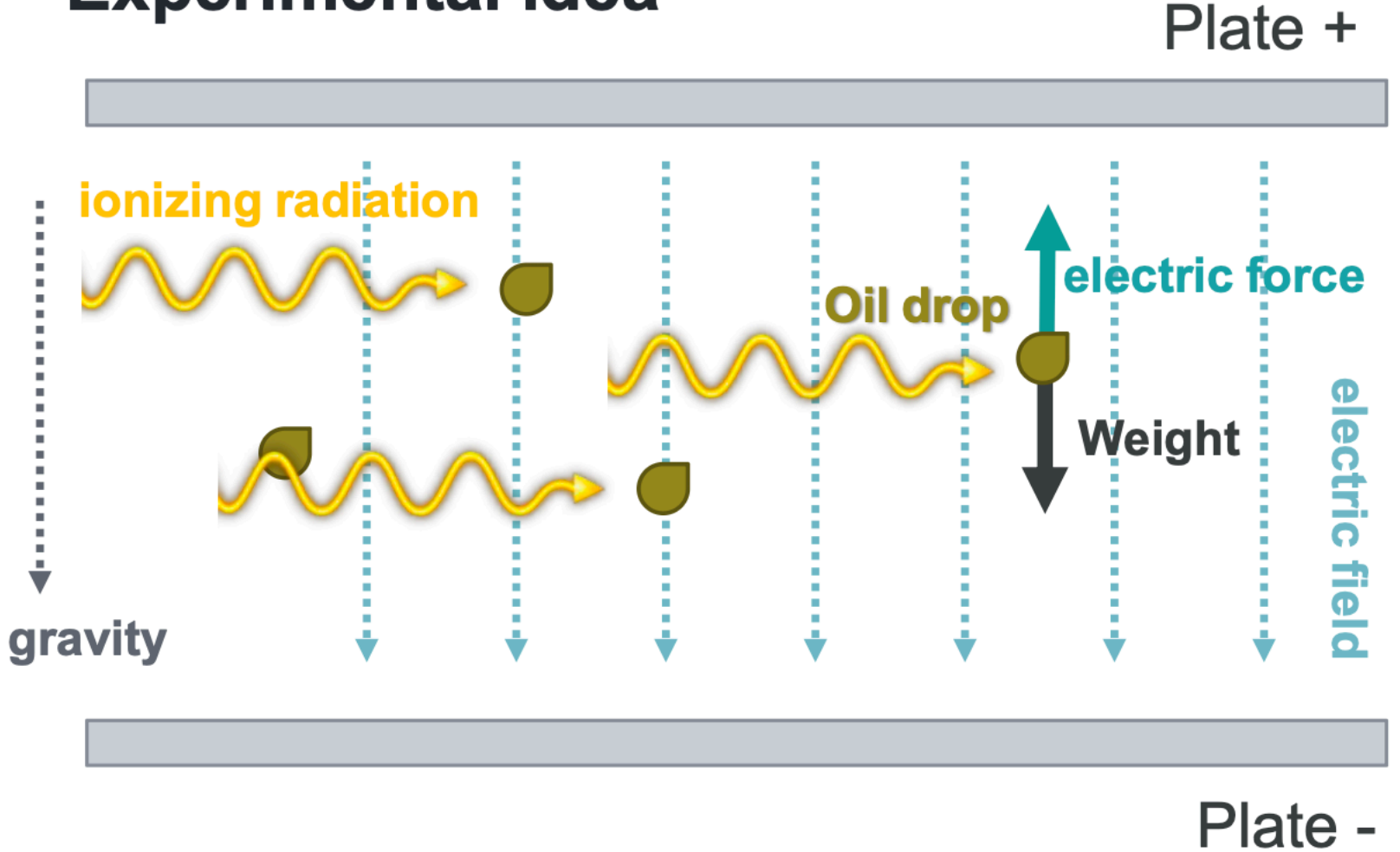

Acc.V Spot Magn Det WD Exp
30.0 kV 3.0 11397x SE 8.0 3
2 µm

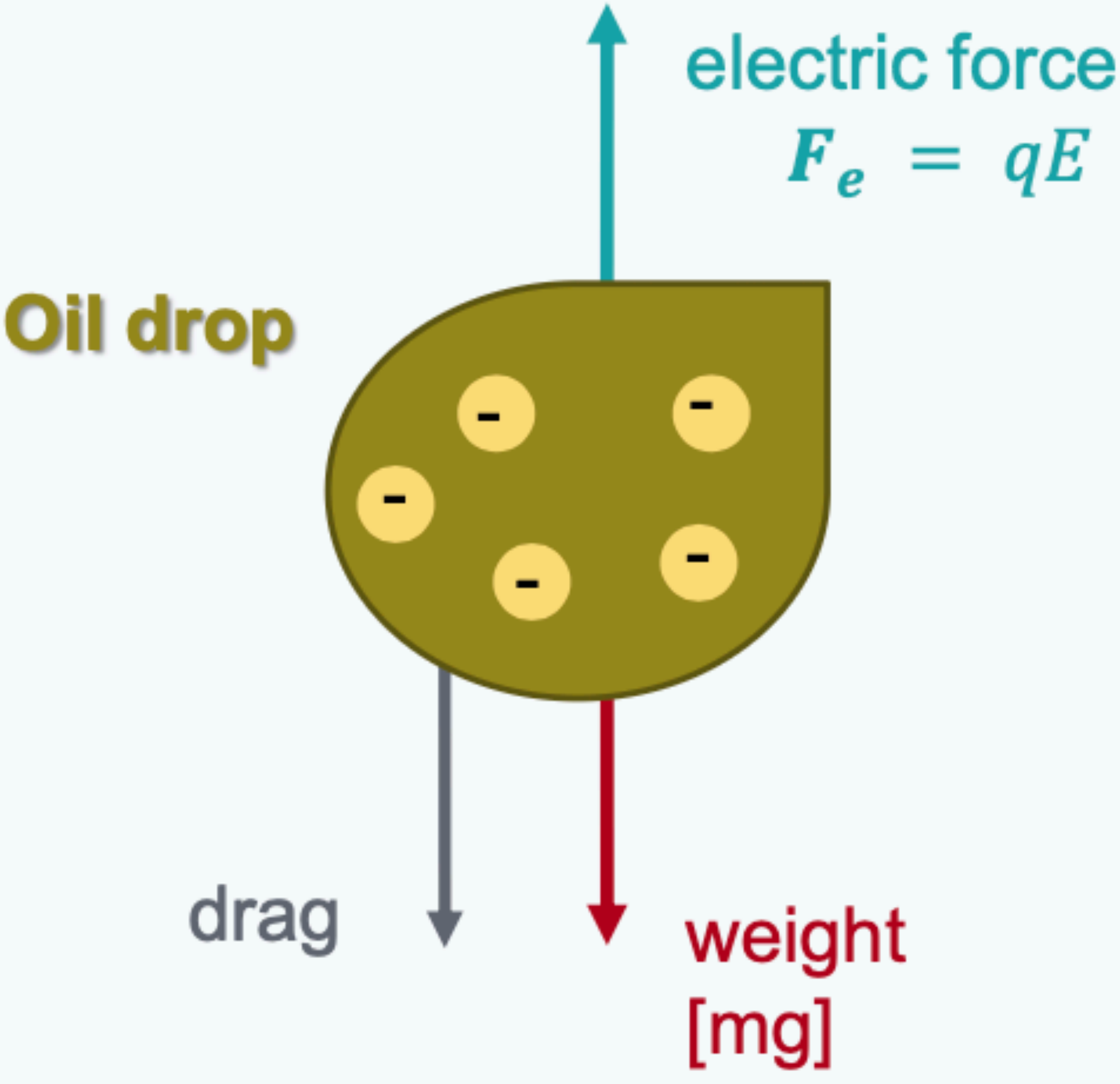

100 nm

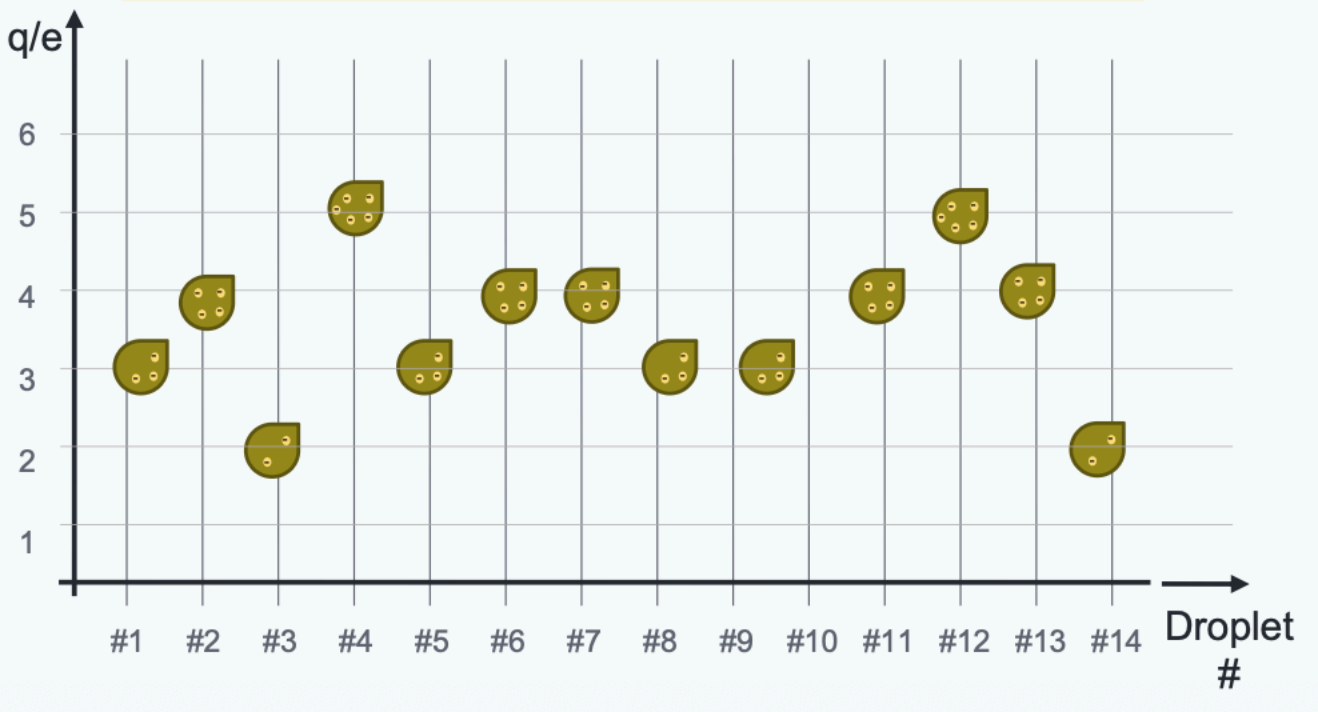

Fiera di Padova
Via Niccolò Tommaseo
Via Luigi Piccinato
Via Ugo Bassi
Via Francesco Rismondo
Via San Leonardo Murialdo
Via del Pesca
Via de
Hotel NH Padova
6 min
1,9 km
Liceo Artistico Statale "Amedeo Modigliani"
Centro Copia Stecchini - Tesi 24 ore
Copisteria
Dipartimento di Matematica "Tullio...
Via Venezia
Via Trieste
Via Leonardo Loredan
Best Western Plus Hotel Galileo
Porta Portello
Lungargine del Piovego
Via Giovanni Gradenigo
Dipartimento di Fisica Galileo Galilei
San Gaetano

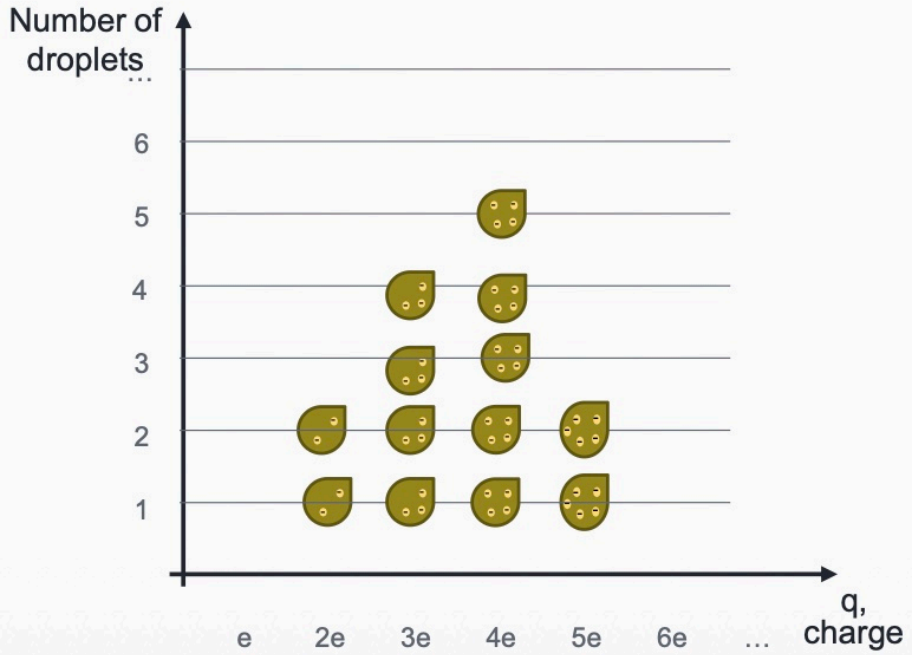


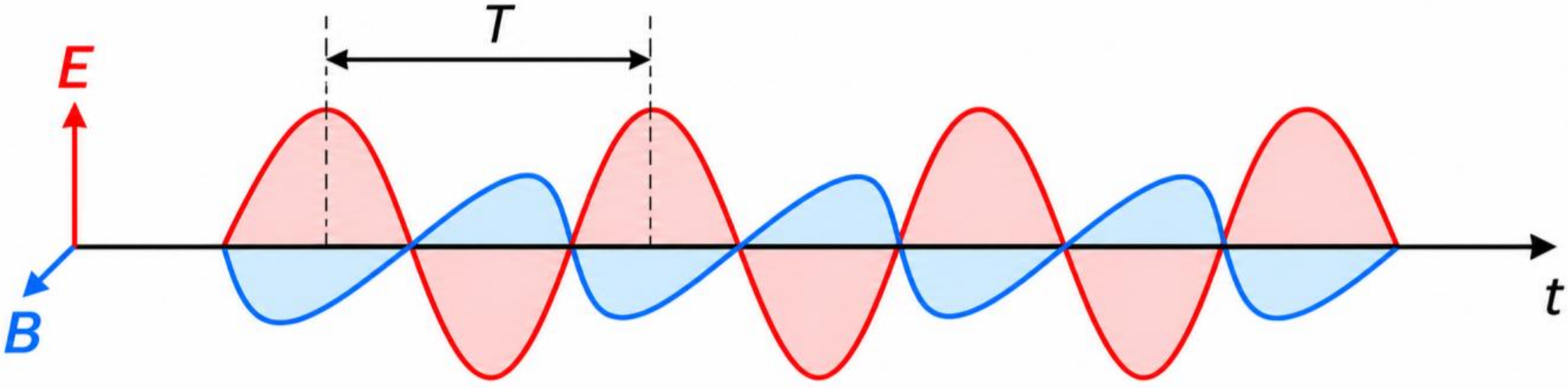
Experimental idea

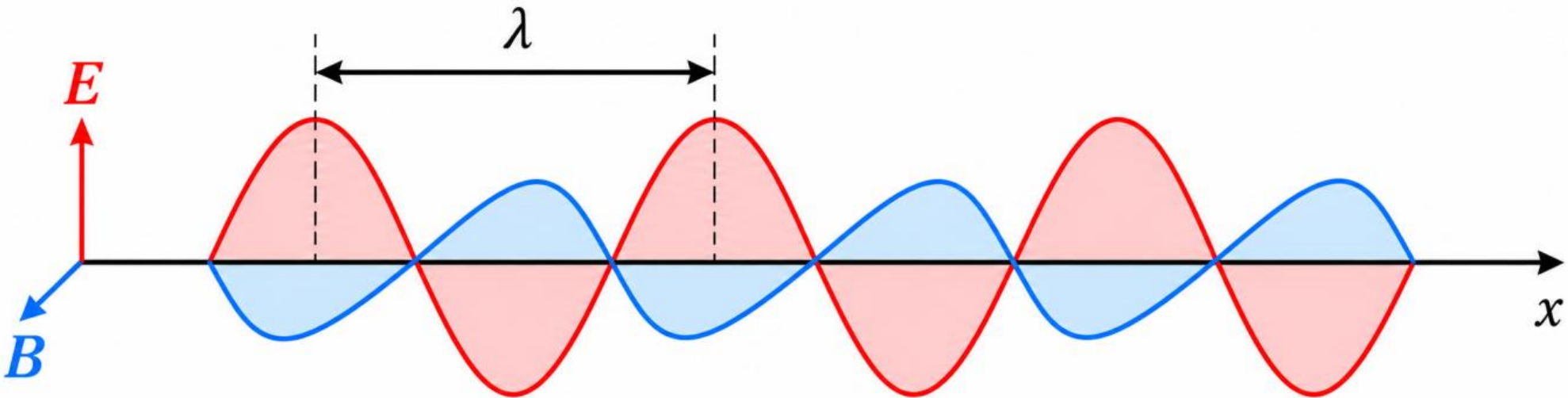


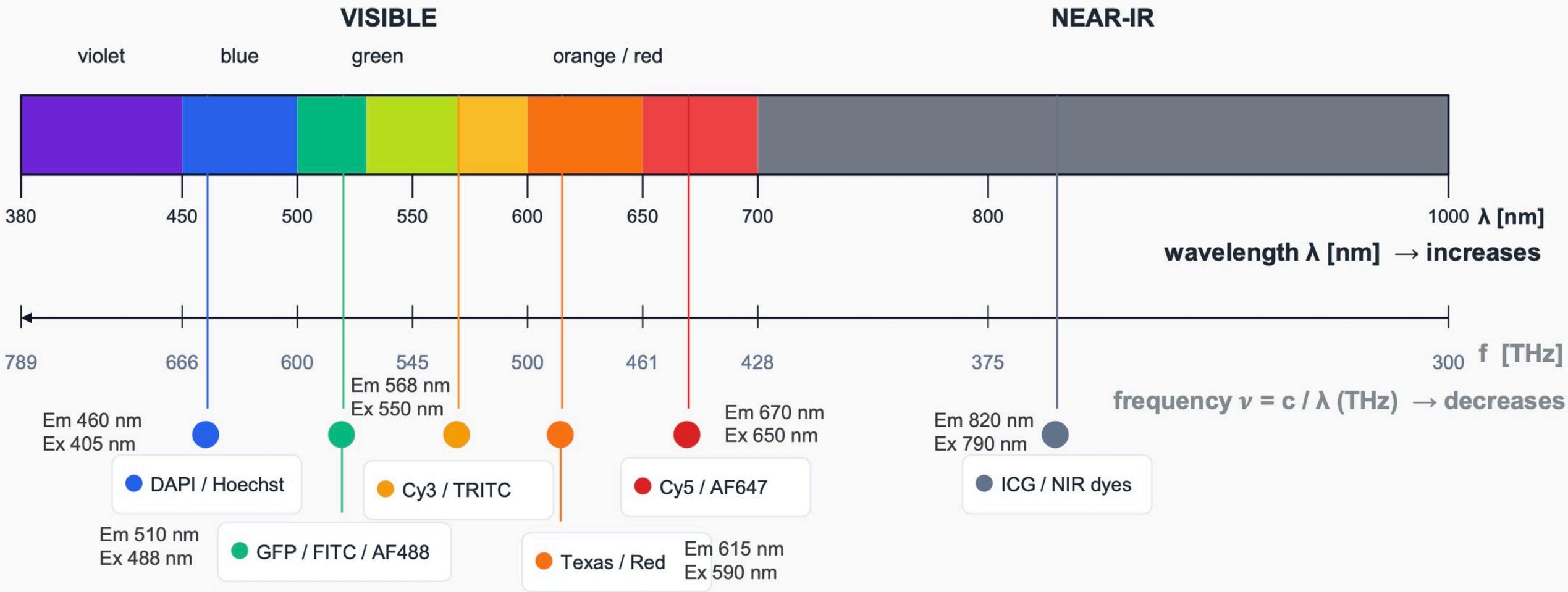

electric force
$F_e = qE$
Oil drop
drag
weight
[mg]

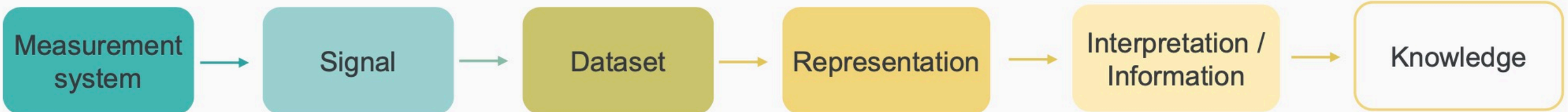

q/e
6
5
4
3
2
1
#1
#2
#3
#4
#5
#6
#7
#8
#9
#10
#11
#12
#13
#14
Droplet #

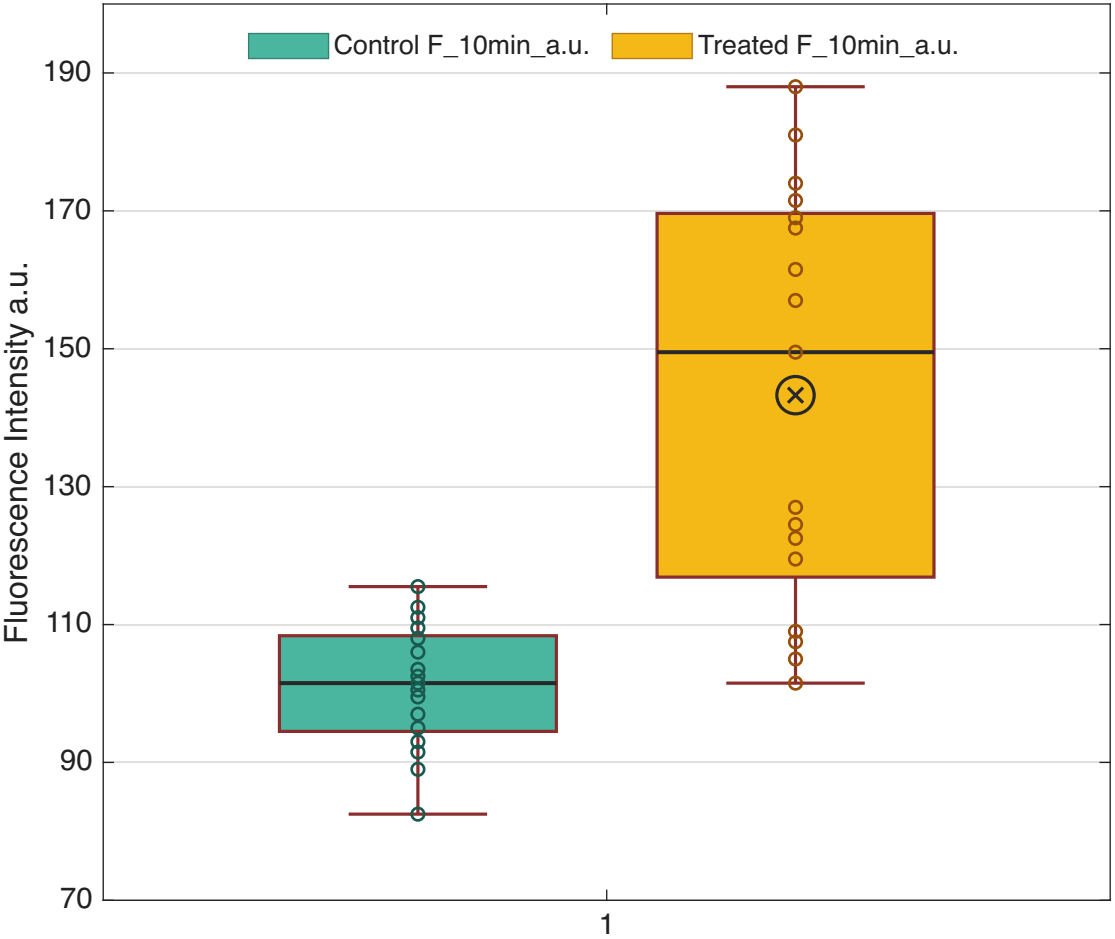

Number of droplets
...
6
5
4
3
2
1
e
2e
3e
4e
5e
6e
...
q, charge

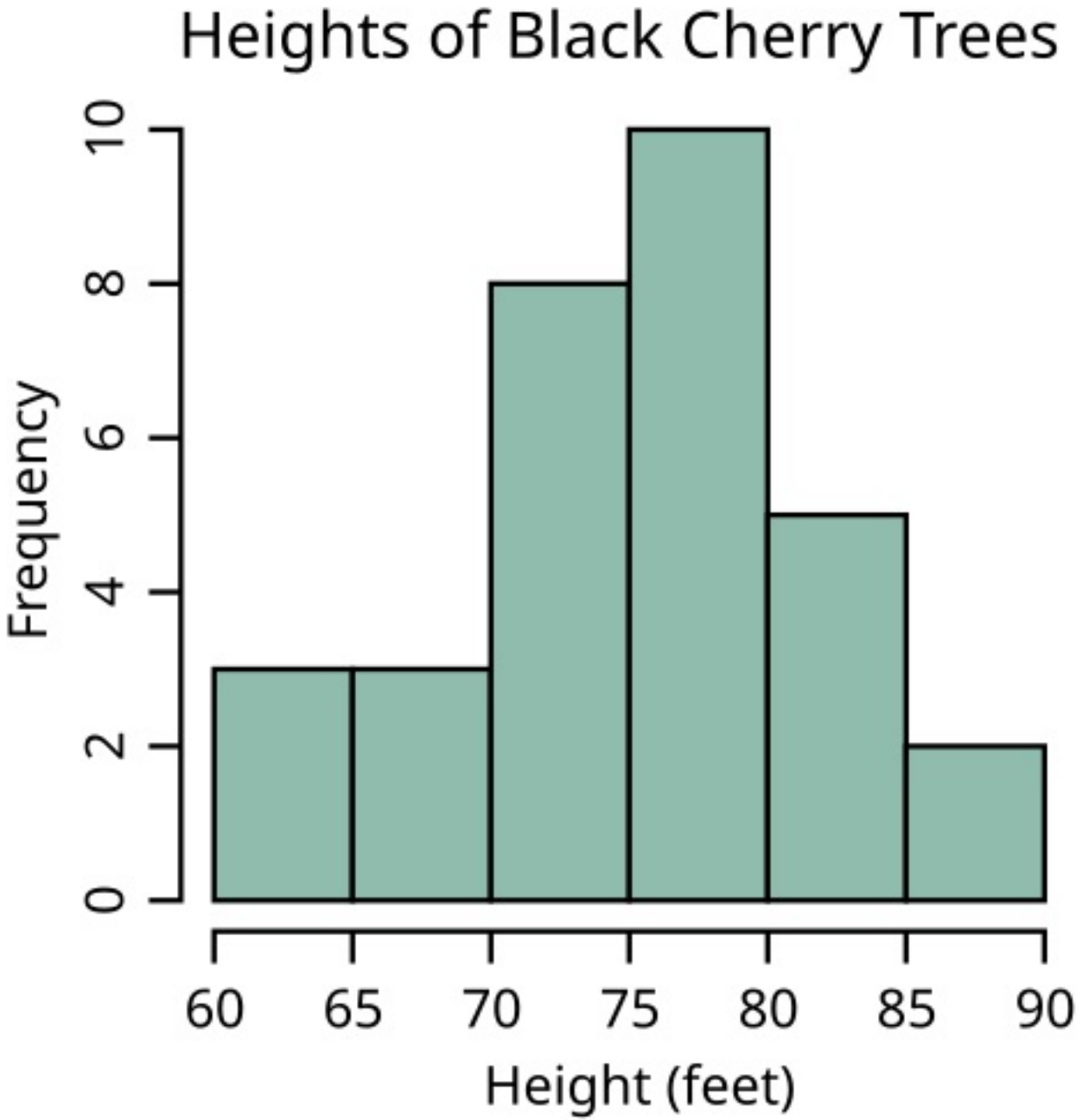

T
E
B
t

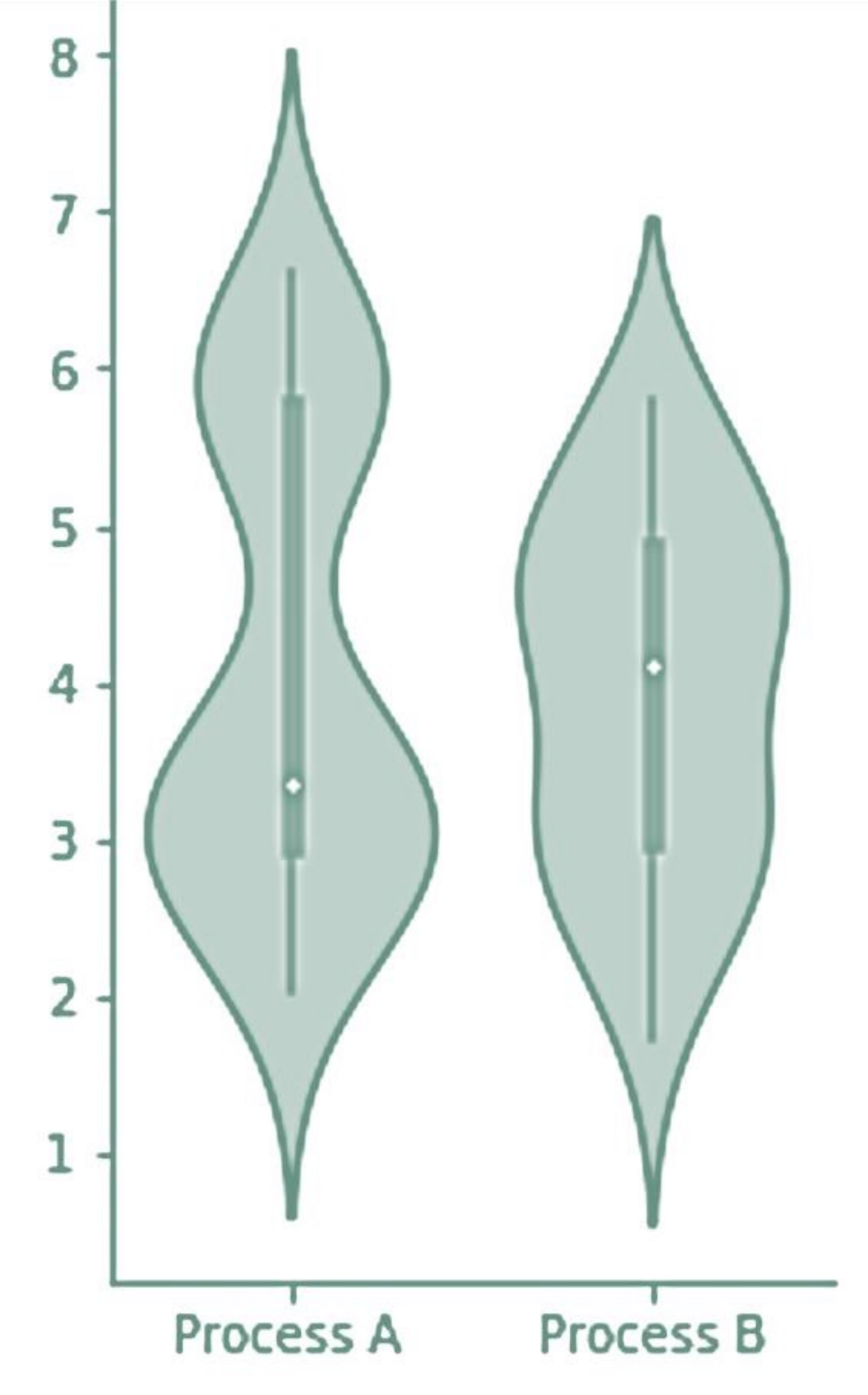

λ
E
B
x

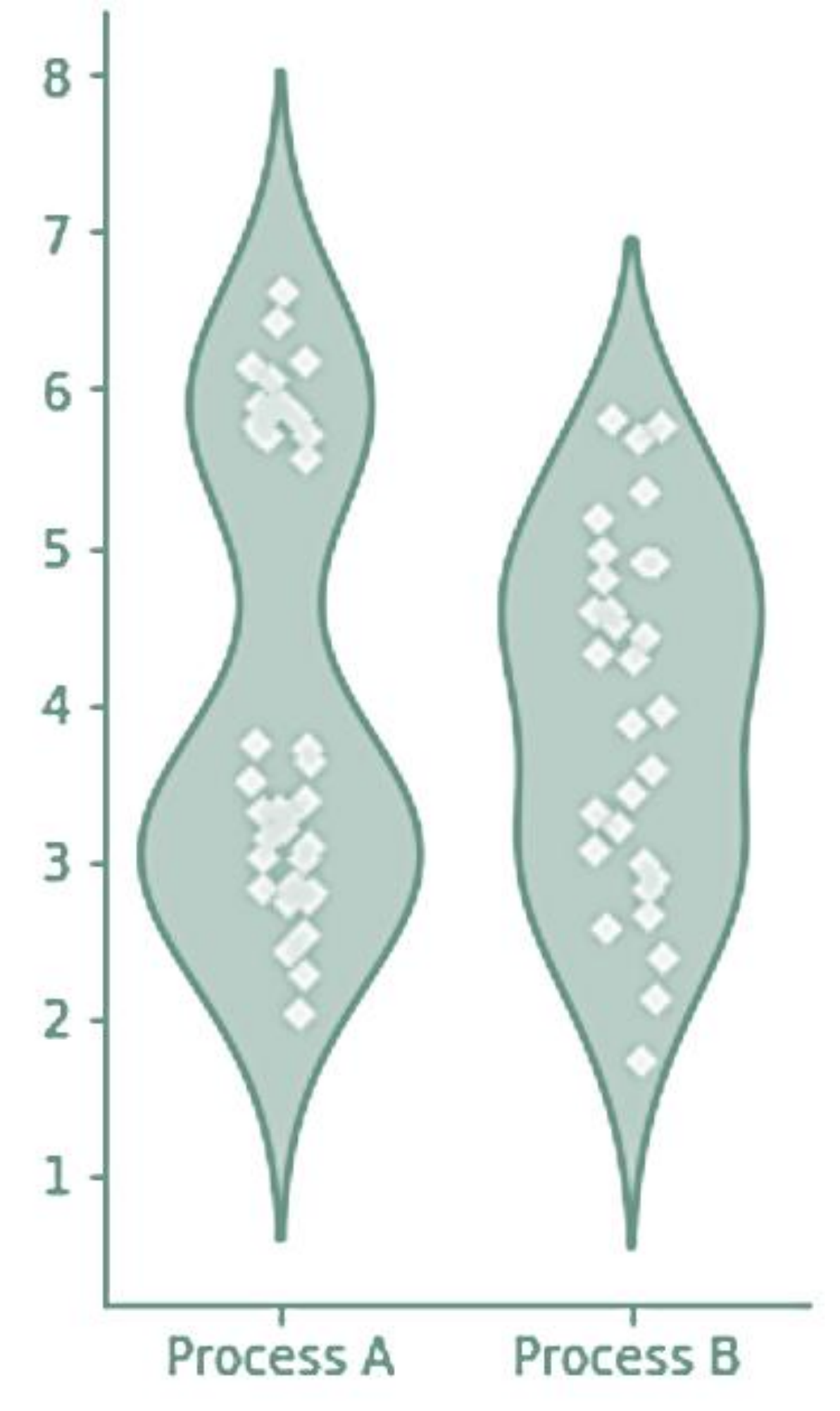

VISIBLE
NEAR-IR
violet
blue
green
orange / red
380
450
500
550
600
650
700
800
1000
λ [nm]
wavelength λ [nm] → increases
789
666
600
545
500
461
428
375
300
f [THz]
frequency ν = c / λ (THz) → decreases
Em 460 nm
Ex 405 nm
DAPI / Hoechst
Em 510 nm
Ex 488 nm
GFP / FITC / AF488
Em 568 nm
Ex 550 nm
Cy3 / TRITC
Texas / Red
Em 615 nm
Ex 590 nm
Em 670 nm
Ex 650 nm
Cy5 / AF647
Em 820 nm
Ex 790 nm
ICG / NIR dyes

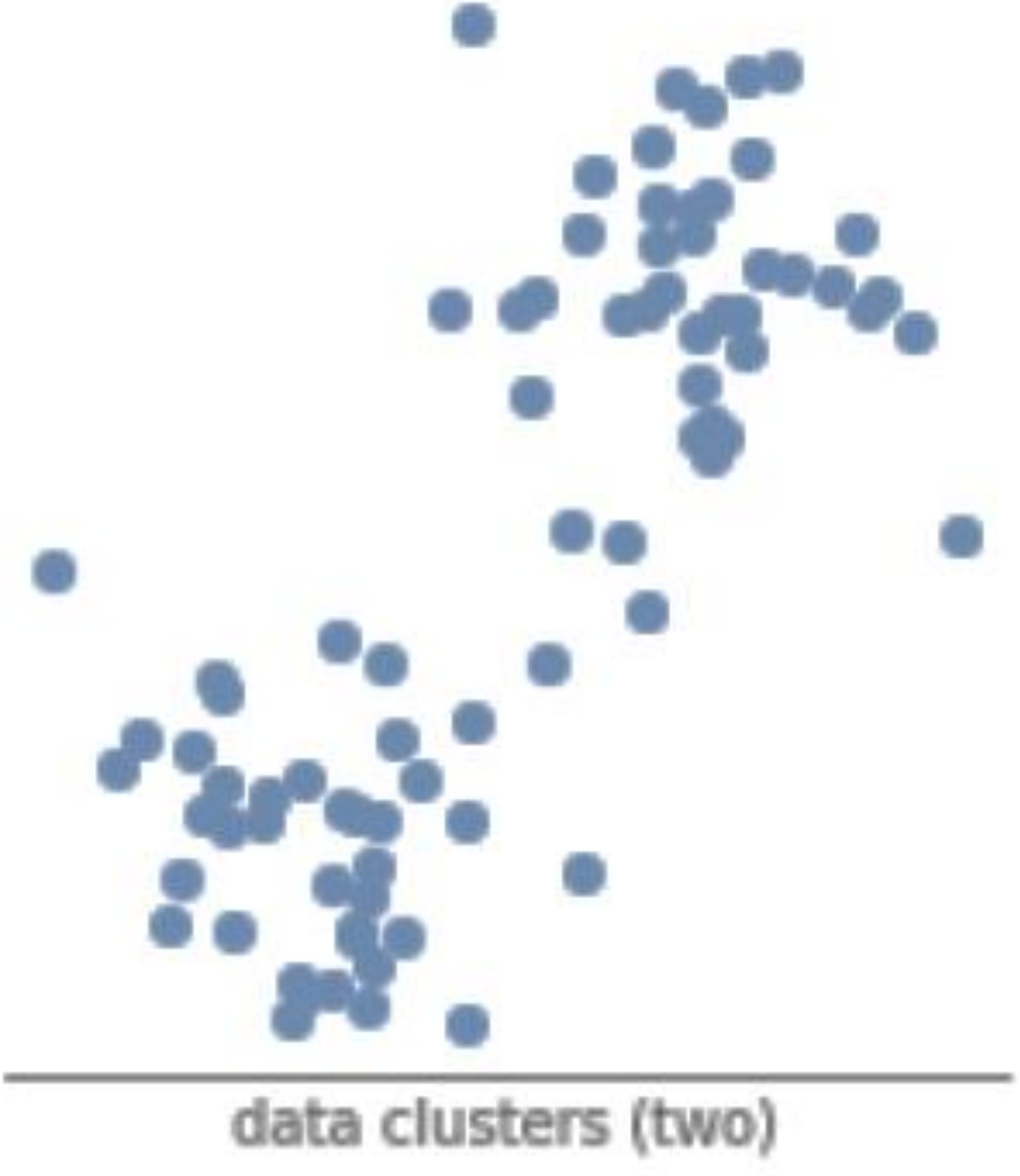

Measurement system
Signal
Dataset
Representation
Interpretation / Information
Knowledge

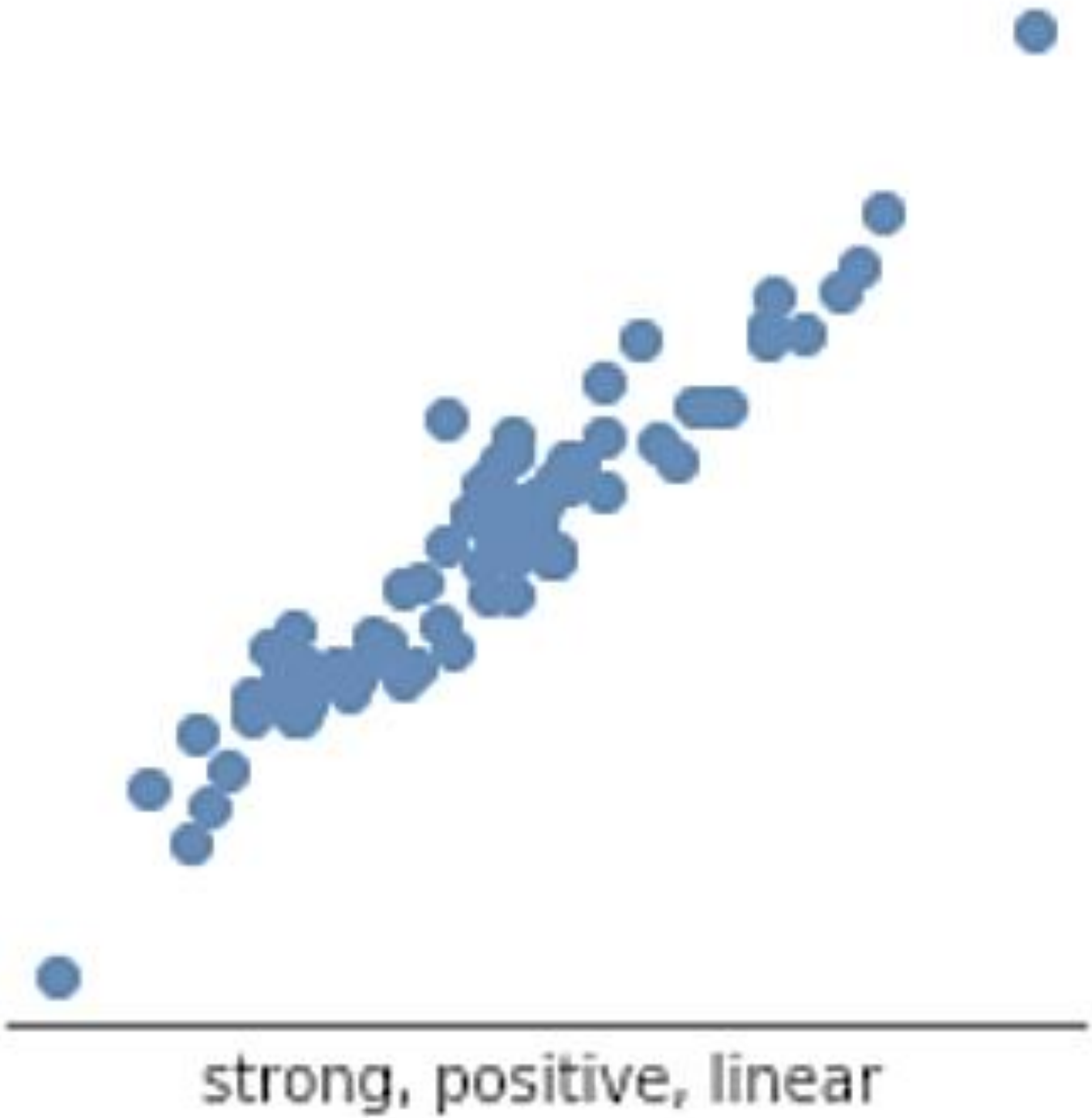

Control F_10min_a.u.
Treated F_10min_a.u.
Fluorescence Intensity a.u.
190
170
150
130
110
90
70
1

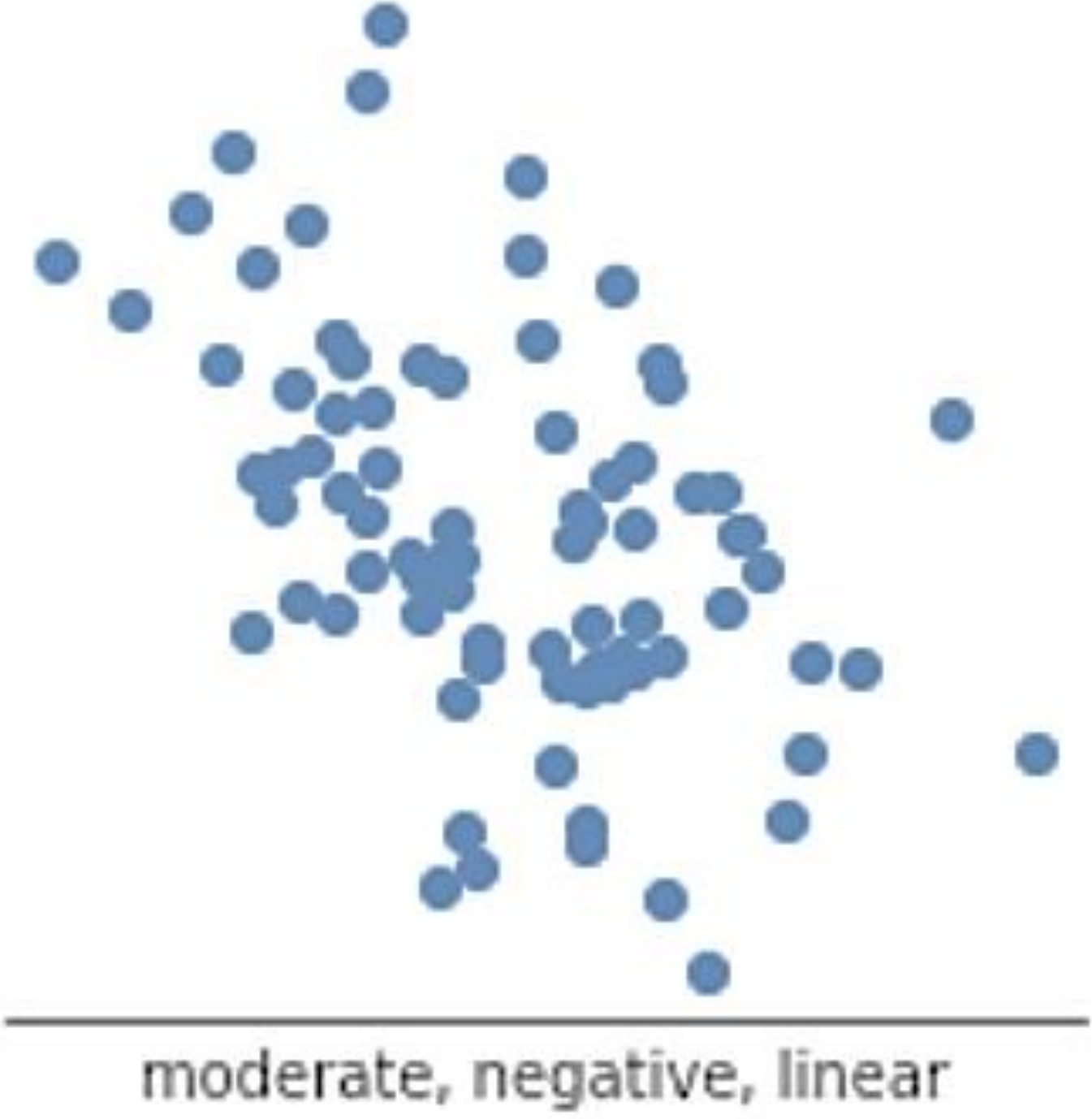

Heights of Black Cherry Trees
Frequency
0
2
4
6
8
10
60
65
70
75
80
85
90
Height (feet)

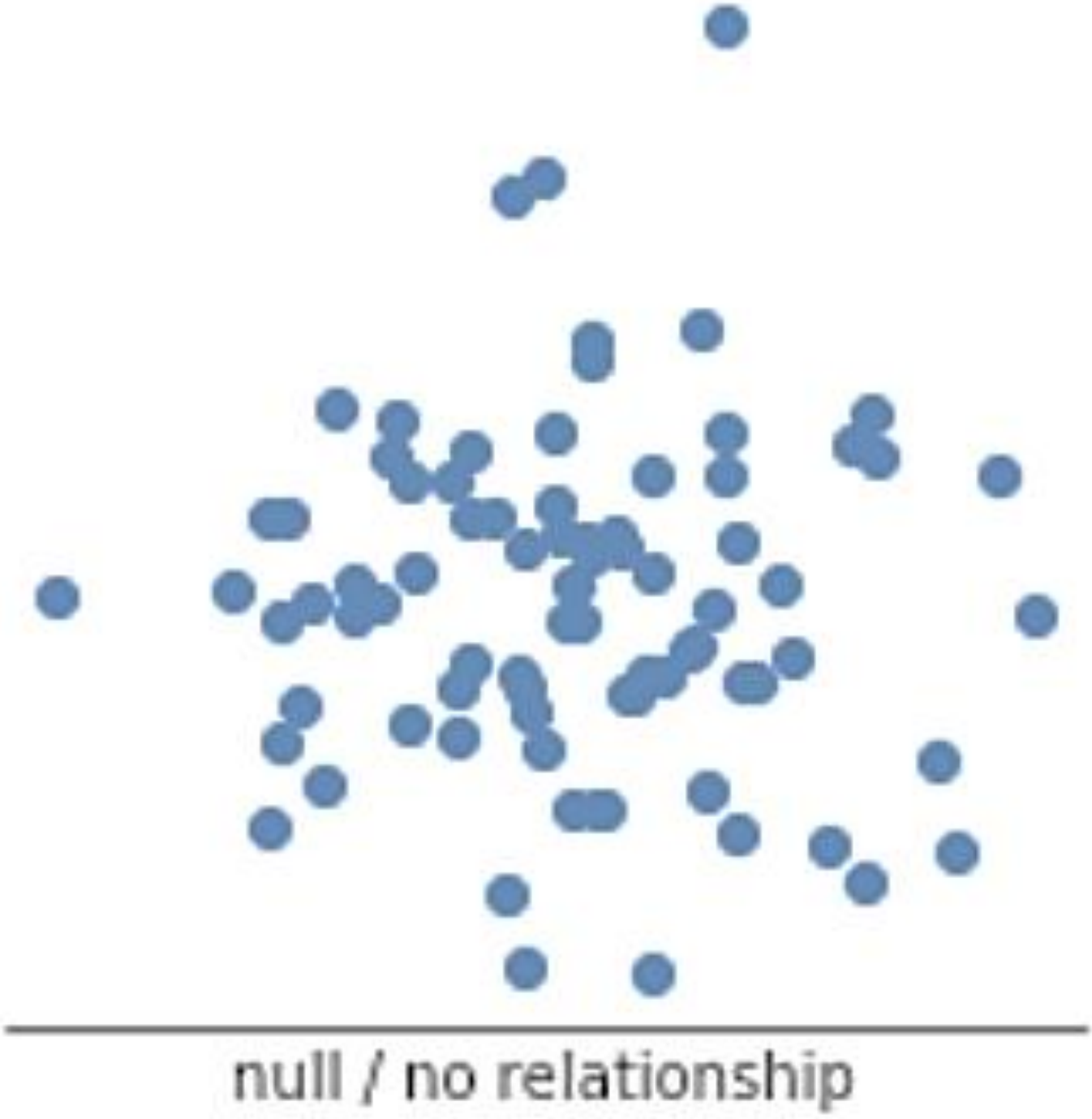

8
7
6
5
4
3
2
1
Process A
Process B

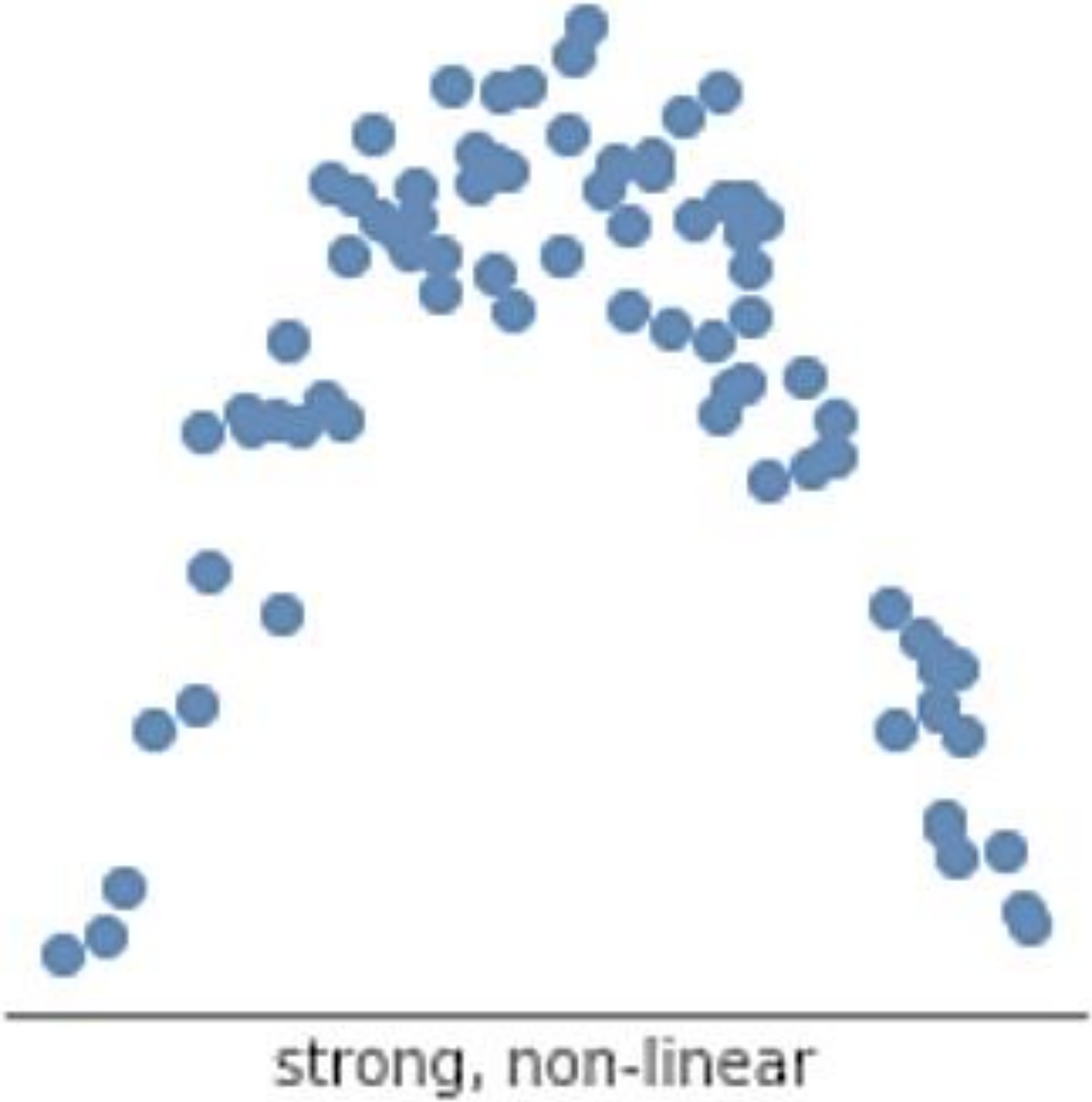

8
7
6
5
4
3
2
1
Process A
Process B


data clusters (two)

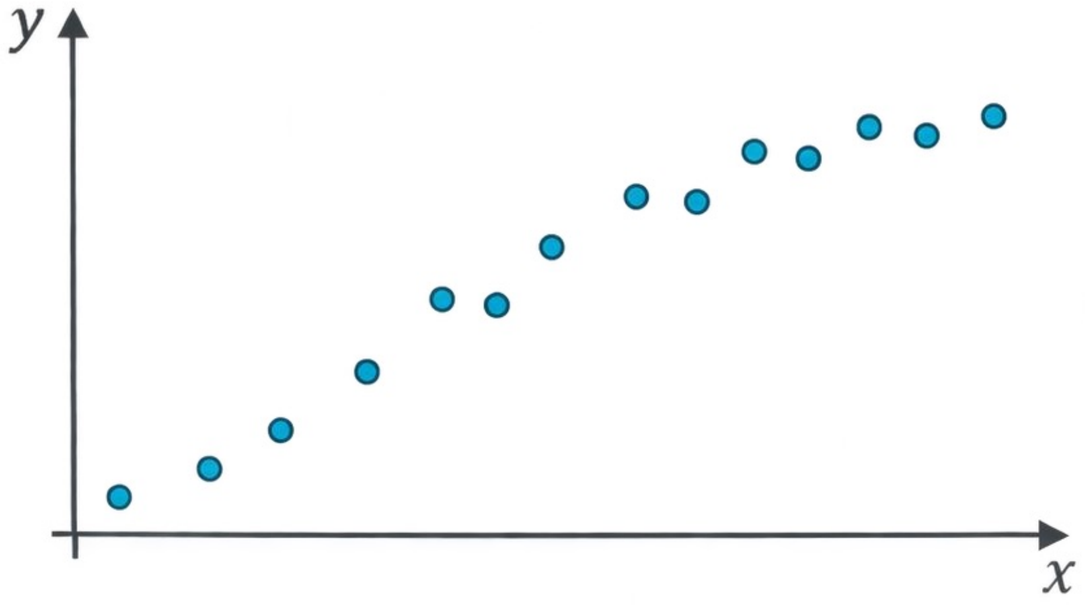

strong, positive, linear


moderate, negative, linear

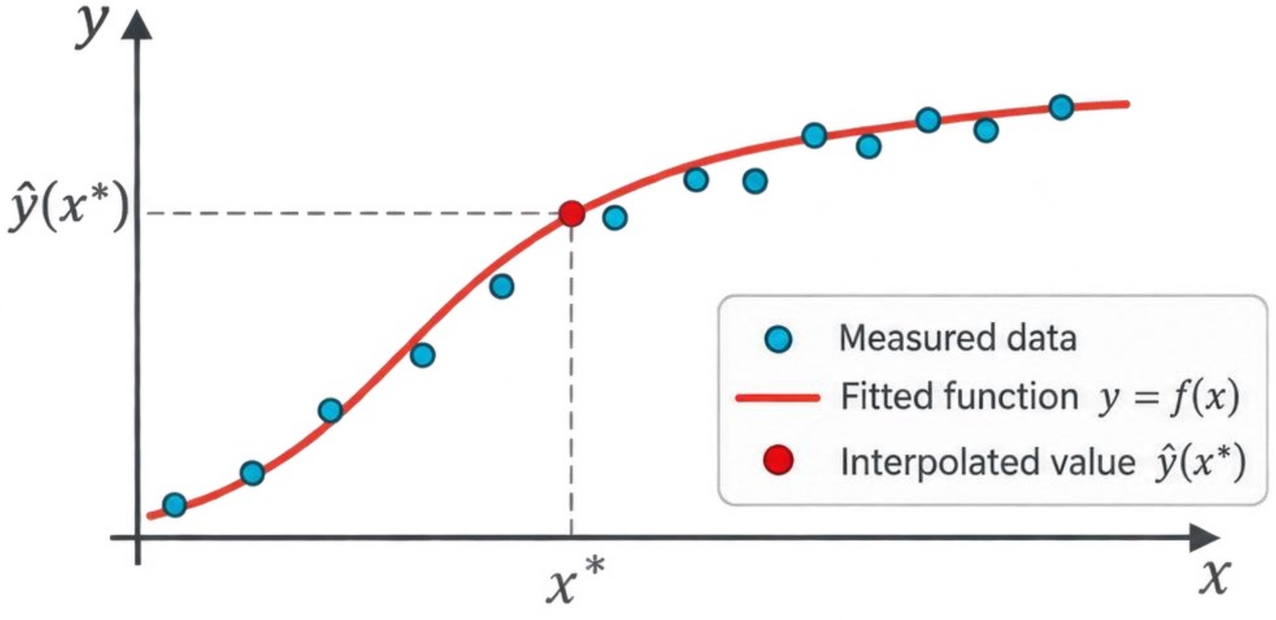

null / no relationship


strong, non-linear

Measured data



Functional approximation

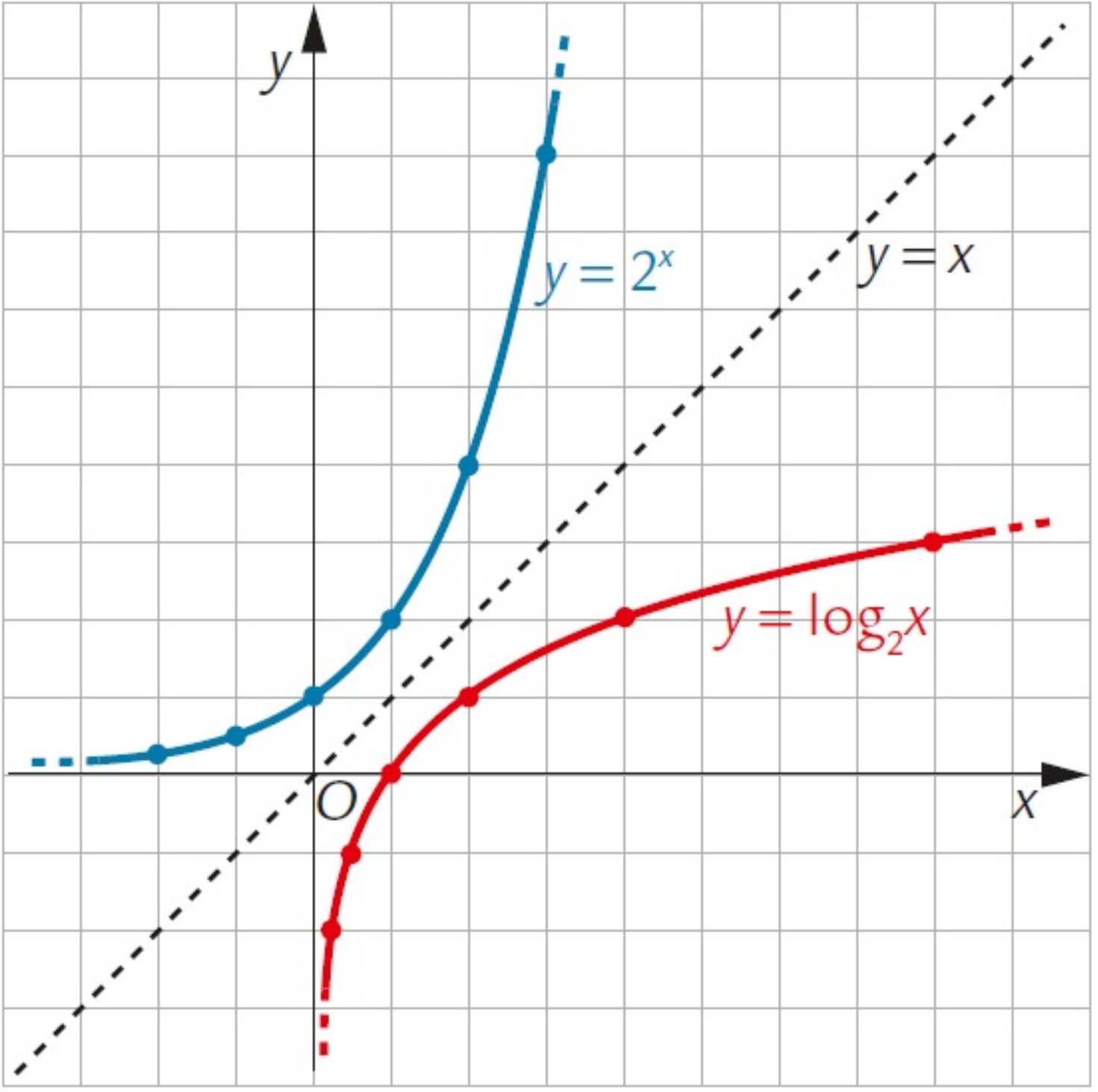


Discrete measurements

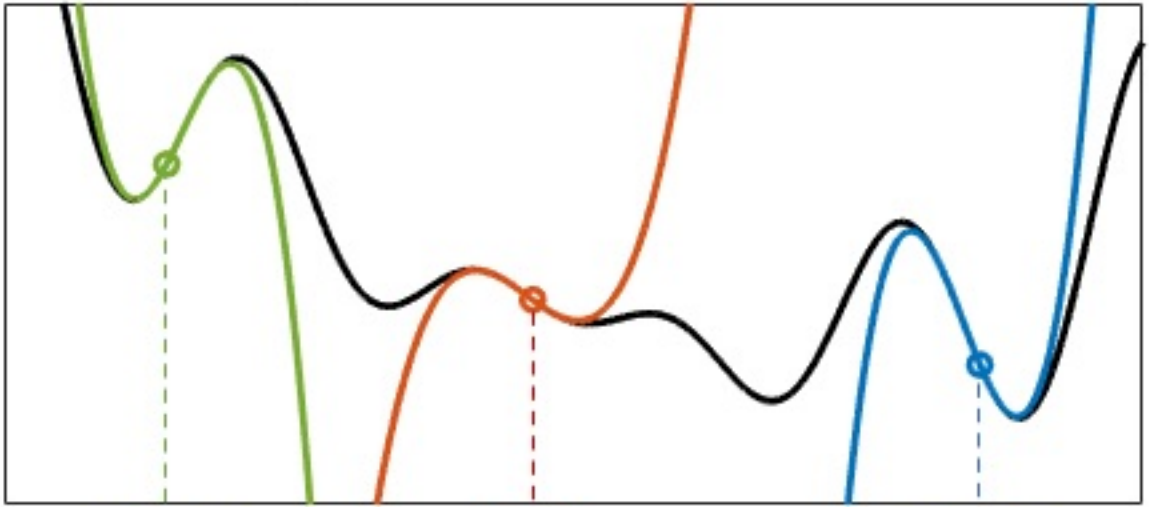
Values of y measured only at specific input points x.

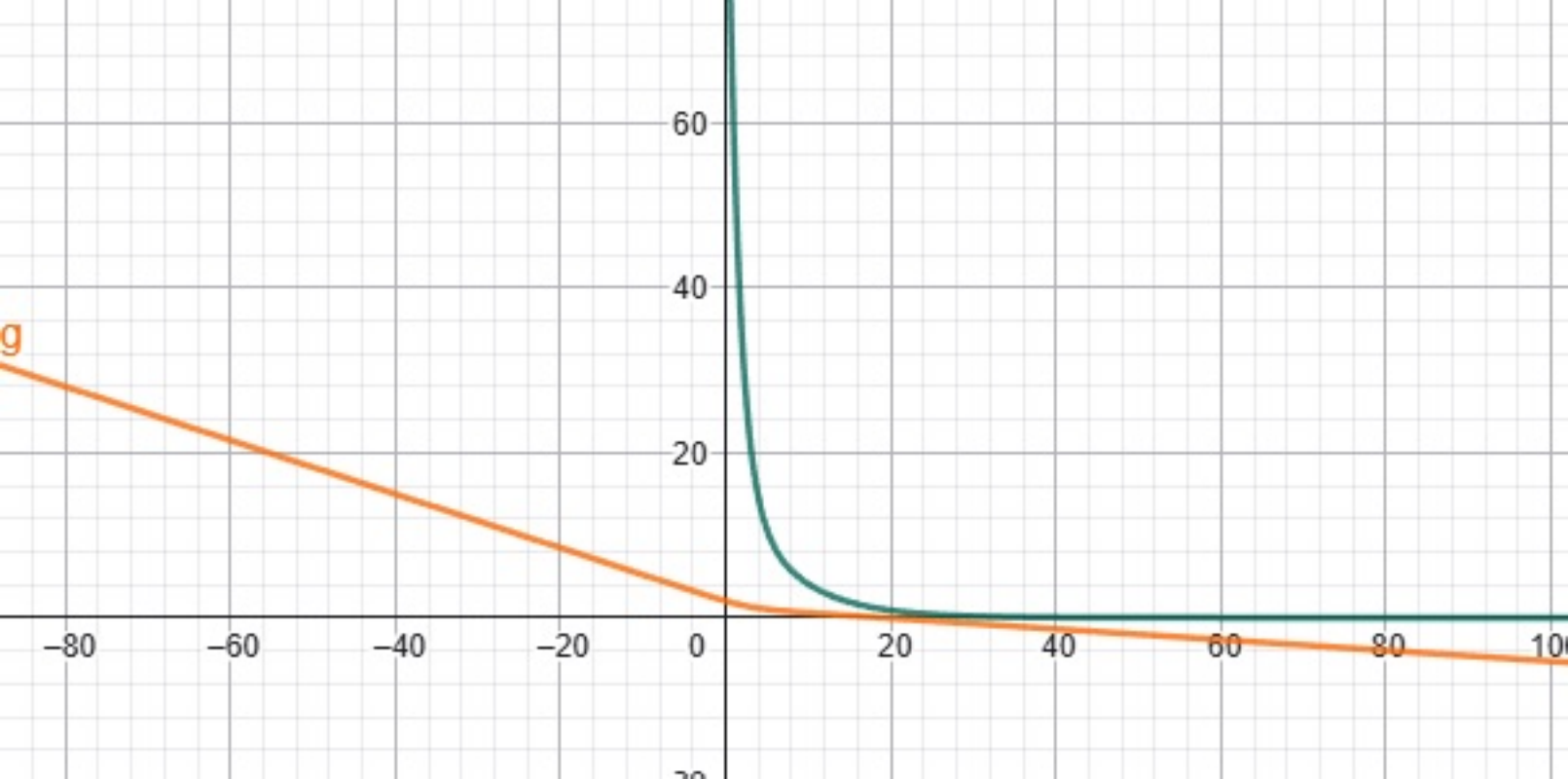
Continuous representation and interpolation

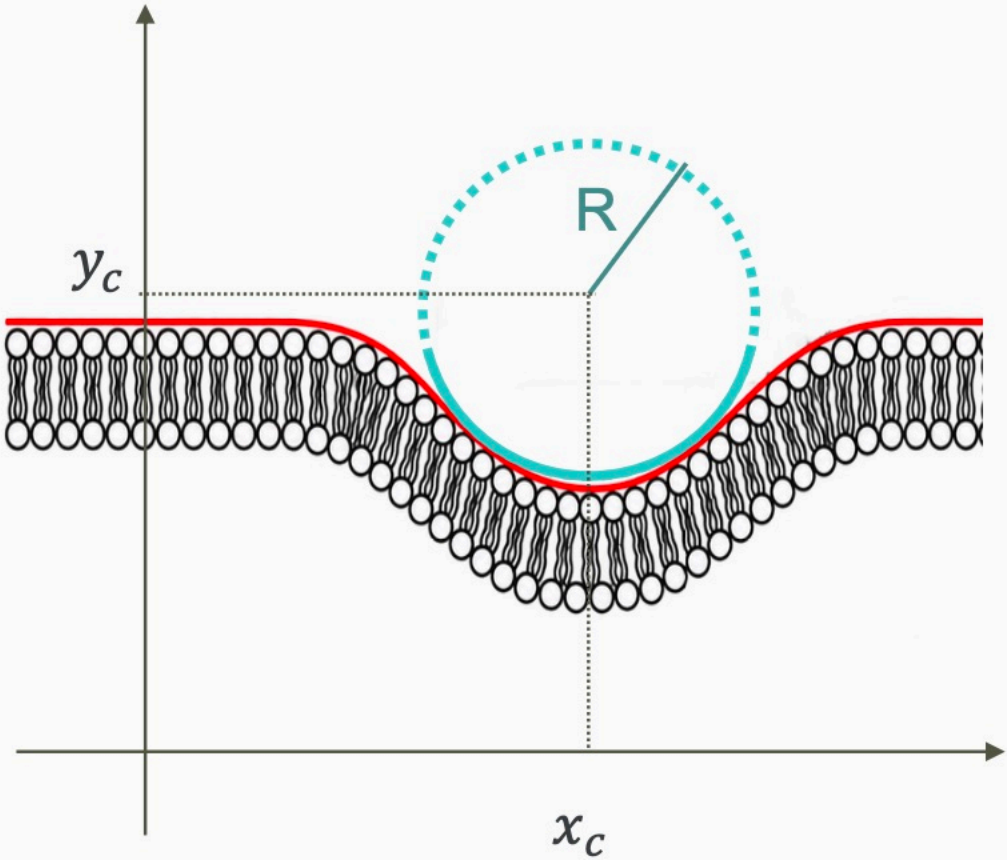
A mathematical function $f(x)$ approximates the data, allowing us to estimate values at unmeasured input points.

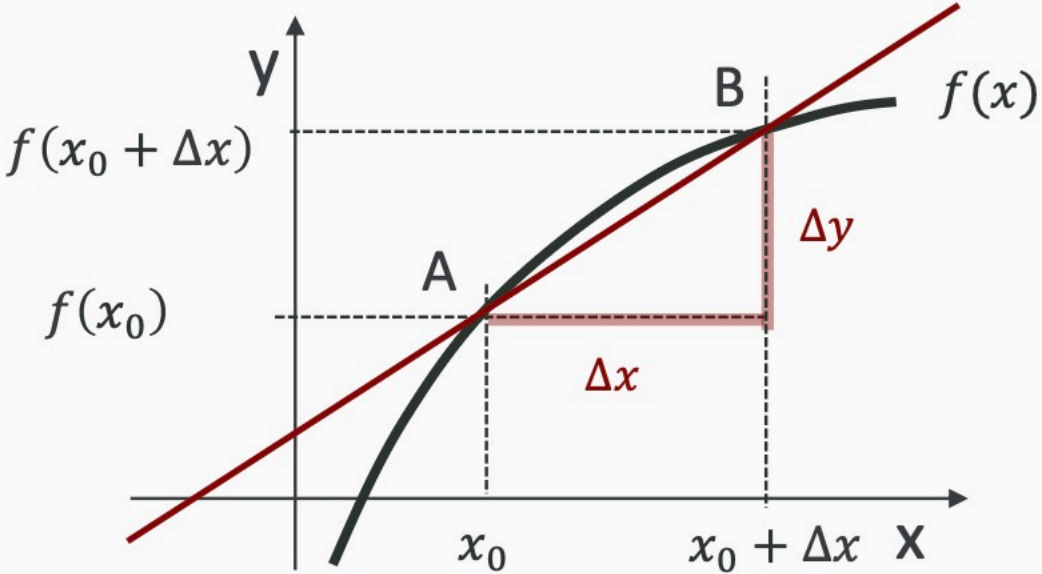

y
x
O
$y = 2^x$
$y = x$
$y = \log_2 x$

Examples with Various Reference Points

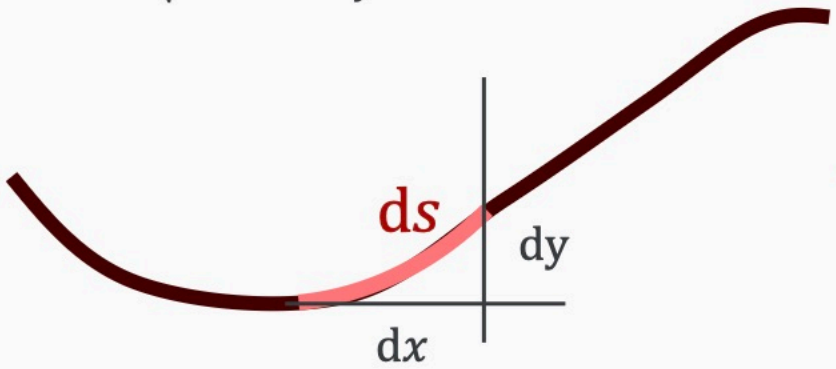


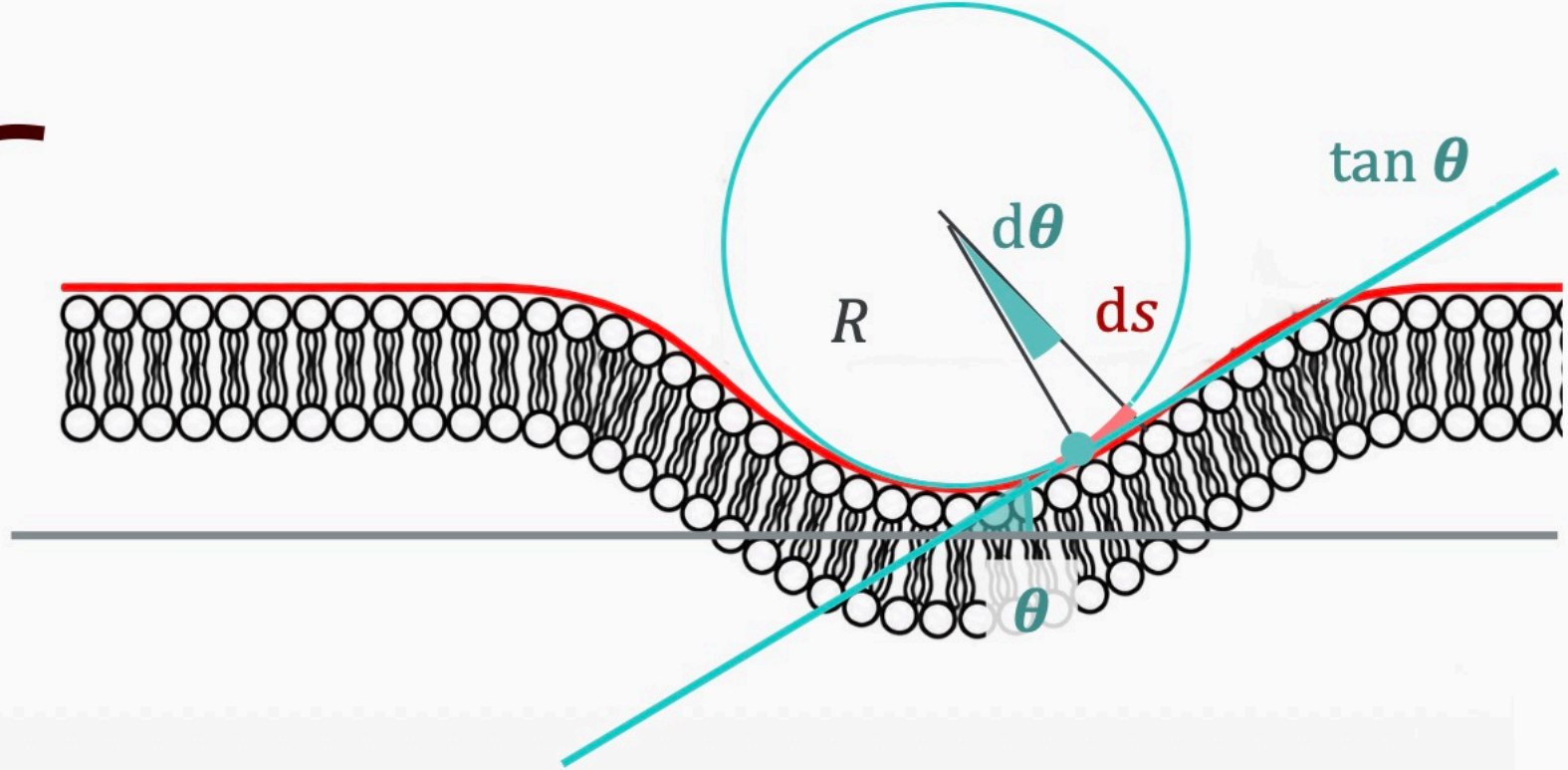

g
60
40
20
−80
−60
−40
−20
0
20
40
60
80

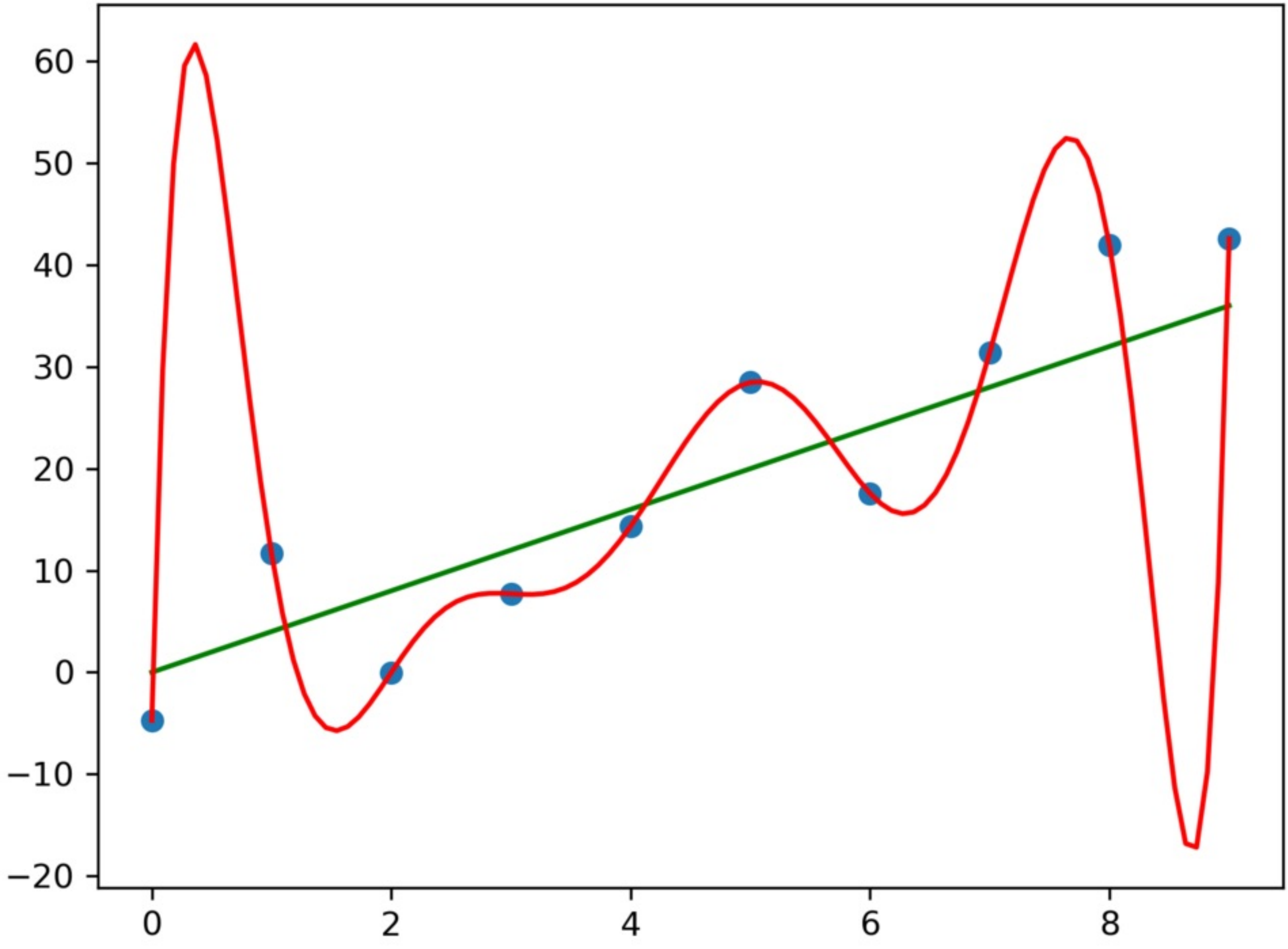

R
y_c
x_c

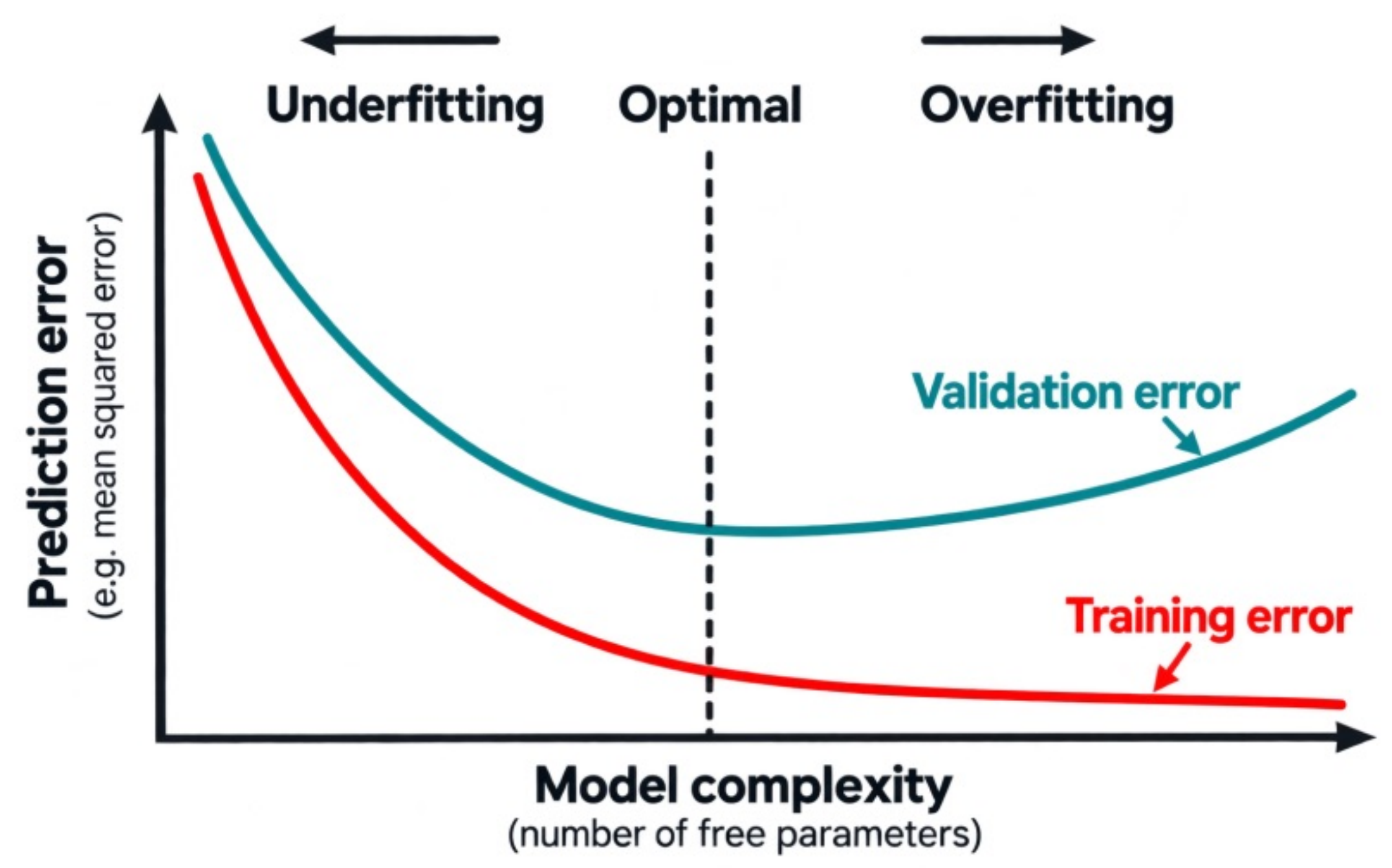

y
B
$f(x)$
$f(x_0 + \Delta x)$
Δy
A
$f(x_0)$
Δx
x_0
$x_0 + \Delta x$
x

$$ds = \sqrt{dx^2 + dy^2}$$






60
50
40
30
20
10
0
−10
−20
0
2
4
6
8


Underfitting
Optimal
Overfitting
Prediction error
(e.g. mean squared error)
Validation error
Training error
Model complexity
(number of free parameters)